Meteor Radar and HamSCI

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MIT Haystack Observatory

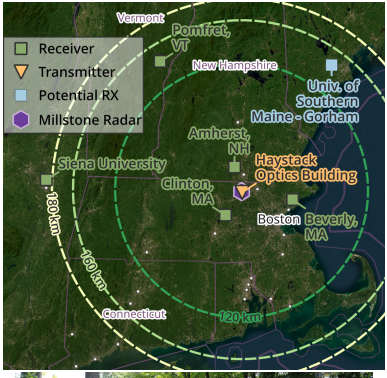
October 21, 2026

HamSCI Meteor Scatter Working Group

SIMONe Haystack

Status

- Calibrated operations since December 2024
- 5-element circular transmitter array
  - Planned restoration of middle 6th antenna
- 5 receiver sites
  - Planned deployment of 1 additional site



Outline

Meteors: Why Do We Care?

Measuring Meteors for Science

Meteor Radars

Meteor Radar Networks

HamSCI

References

Meteors are cool!

Meteoroid input to upper atmosphere

Mass

- On the order of 10 kg/s

Metals

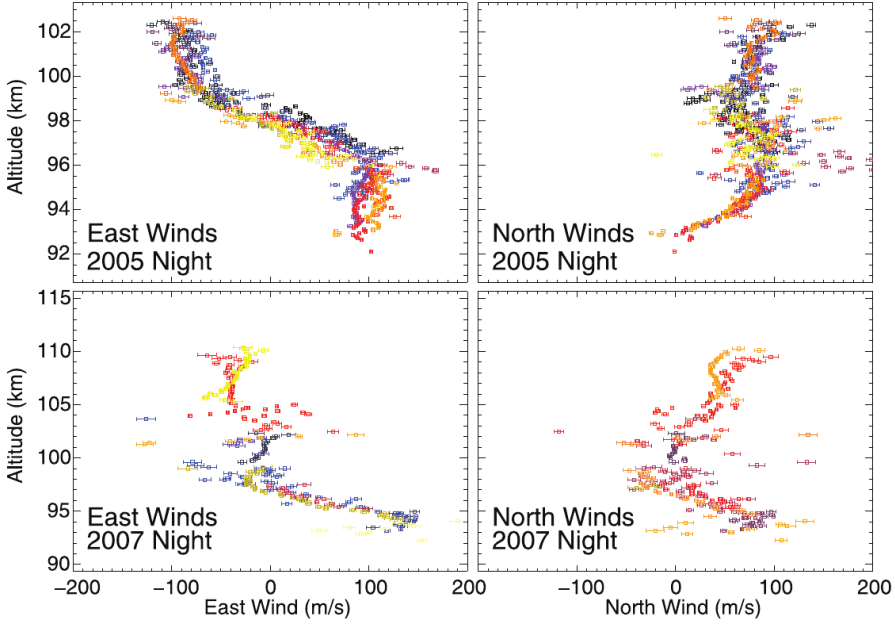
Deposited in mesosphere/lower thermosphere

- Lidar/radar scattering
- Ice condensation nuclei for noctilucent clouds
- Chemistry

Plasma

- Sporadic E
- Instabilities are triggered

Window to geophysical processes



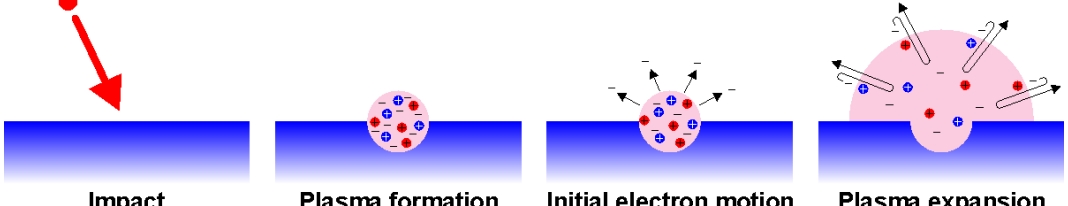
Impacts on satellites

Impact damage to Space Shuttle Atlantis

High speed impacts

Threat to satellites and spacecraft from

- mechanical damage
- electrical damage



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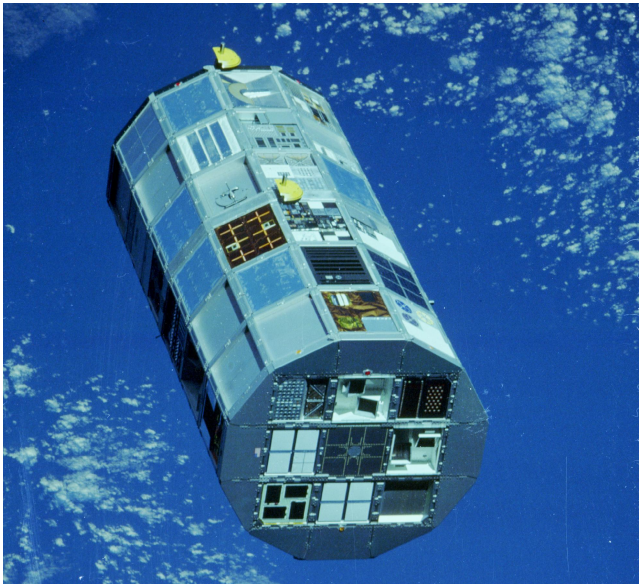
In-situ measurements

LDEF (Long duration exposure facility)



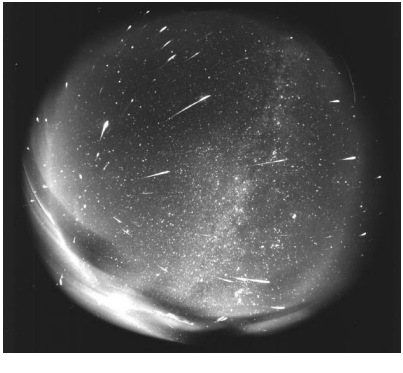
Optical measurements

Crowd sourcing

- Volunteers count the meteors they see

All-sky camera

- Networks across world
- Brightness -> mass
- Velocity



Advantages

Full sky view

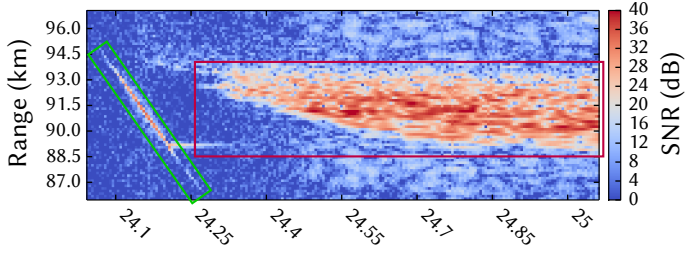
Limitations

Night only
Big meteoroids only

High-power large-aperture radar measurements

Types of scattering

- Head: ball of plasma traveling with the meteoroid
- Trail: plasma left in wake of meteoroid



Advantages

Can see smallest meteors
Day and night

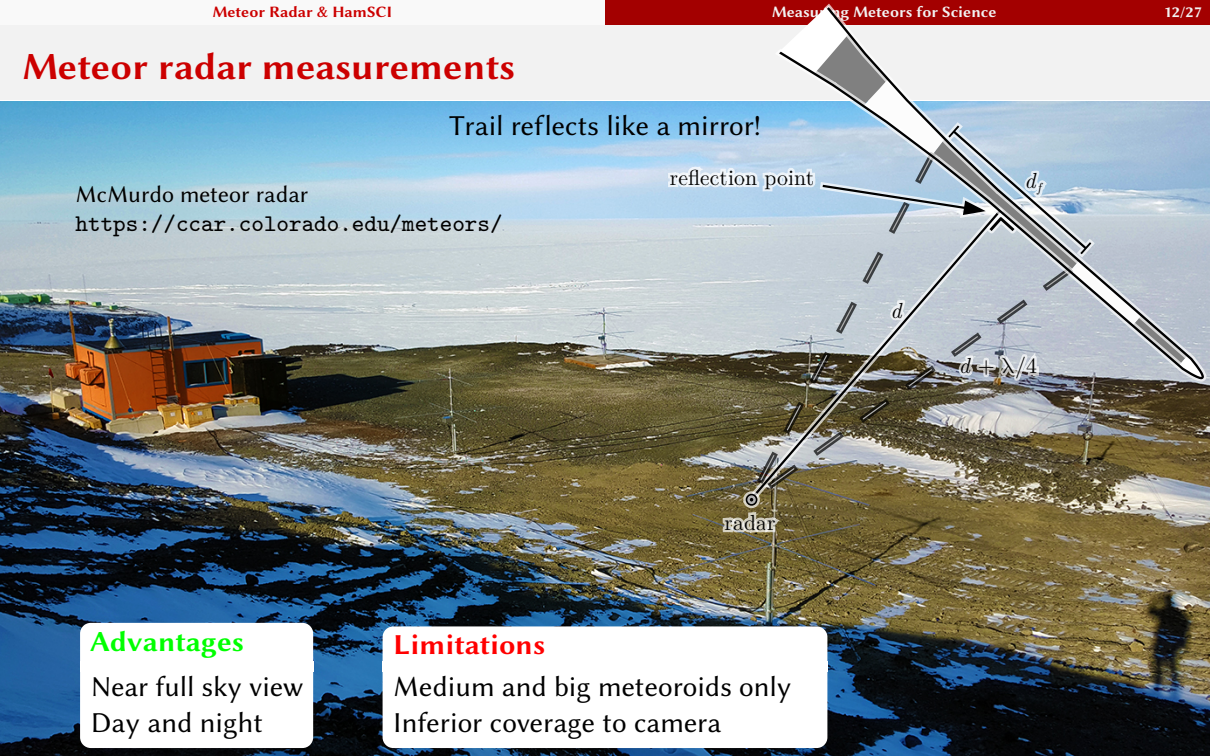
Limitations

Coverage only within small beam
Few HPLA radars, not routine

Meteor radar measurements

Trail reflects like a mirror!

McMurdo meteor radar
https://ccar.colorado.edu/meteors/



Advantages

Near full sky view
Day and night

Limitations

Medium and big meteoroids only
Inferior coverage to camera

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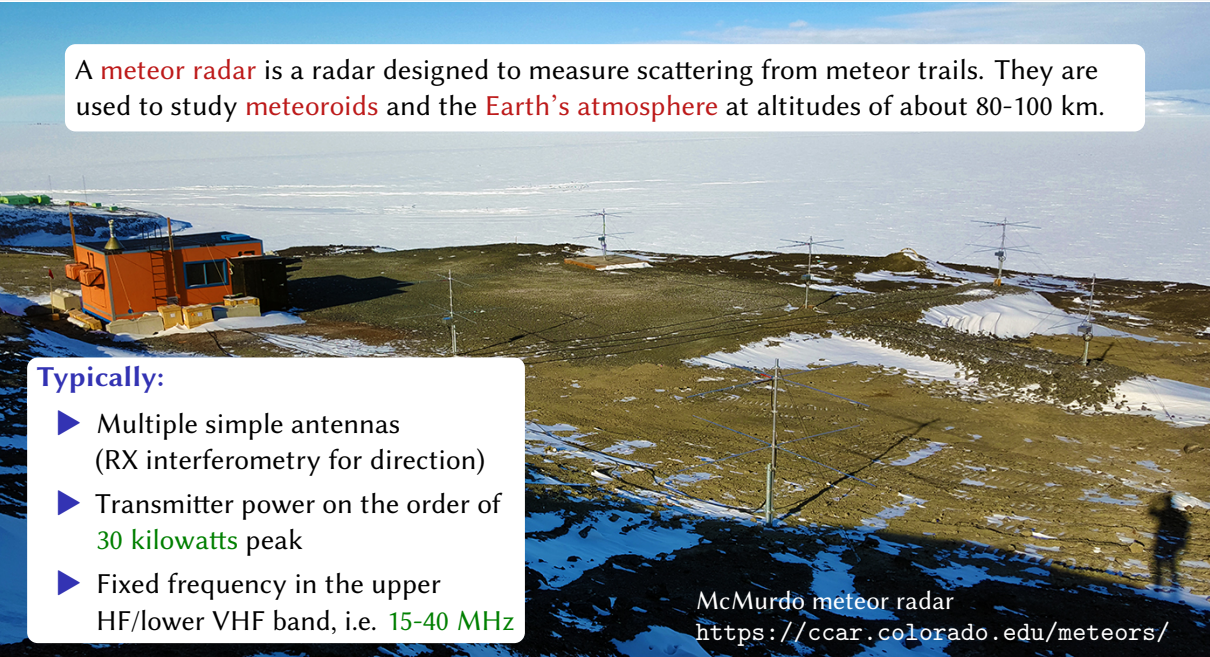
Meteor radar overview

A meteor radar is a radar designed to measure scattering from meteor trails. They are used to study meteoroids and the Earth's atmosphere at altitudes of about 80-100 km.



Typically:

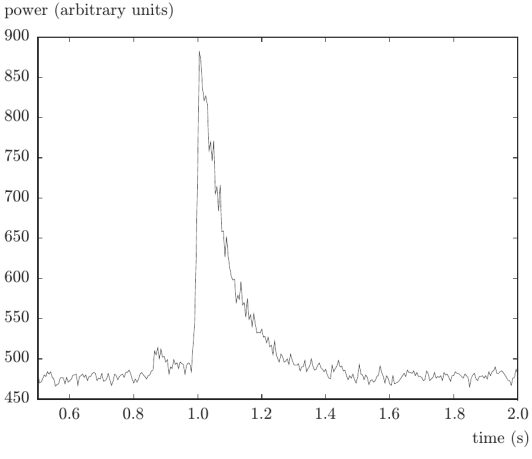
- Multiple simple antennas (RX interferometry for direction)
- Transmitter power on the order of 30 kilowatts peak
- Fixed frequency in the upper HF/lower VHF band, i.e. 15-40 MHz

McMurdo meteor radar
https://ccar.colorado.edu/meteors/

Meteor echoes

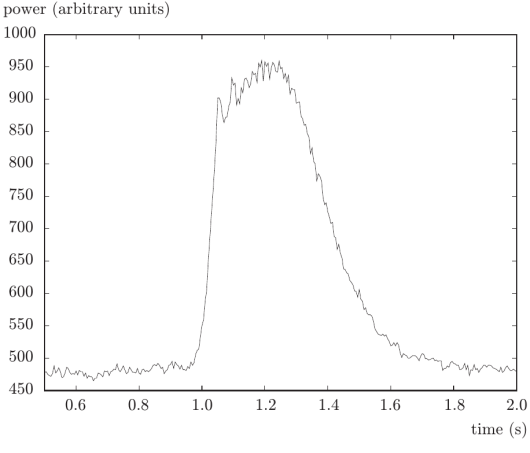
Underdense echo

- Trail plasma frequency less than radar frequency: $f_p \ll f_c$



Overdense echo

- Trail plasma frequency greater than radar frequency: $f_p \gg f_c$

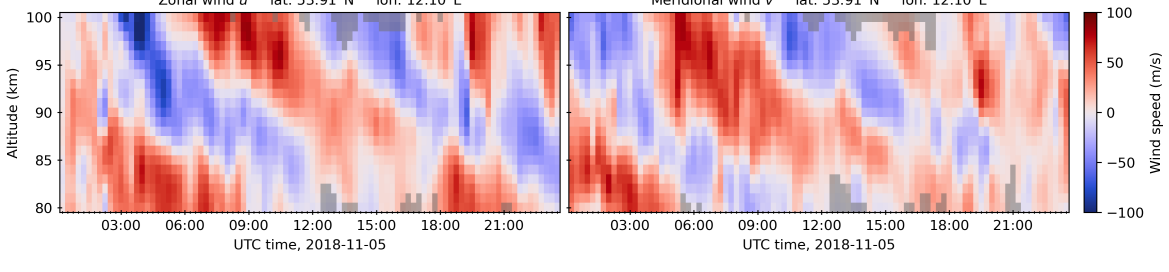


Meteor wind measurements

- Trail plasma drifts with neutral wind
- Movement imparts a Doppler frequency shift in the Bragg vector direction
- Can relate the measured Doppler shift to the wind at the meteor location/time:

$$f_D(x,y,z,t) = \frac{1}{2\pi}\begin{bmatrix} k_x & k_y & k_z \end{bmatrix}\begin{bmatrix} u(x,y,z,t) \\ v(x,y,z,t) \\ w(x,y,z,t) \end{bmatrix}$$

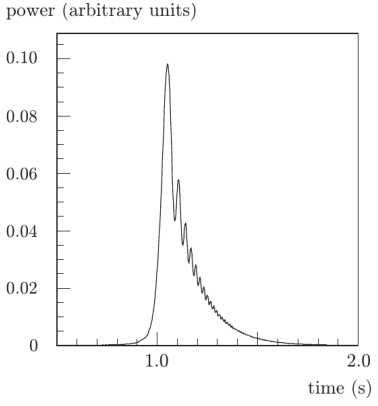
  - k_x, k_y, k_z are Bragg vector components; u, v, and w are the unknown wind components
- Fitting a diverse set of measurements within a given volume:



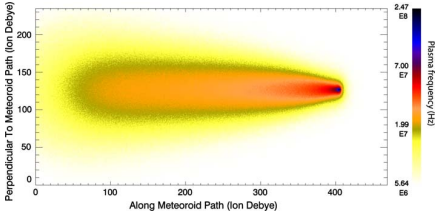
Other useful measurements

Temperature and pressure

- For an underdense echo, the returned power is a function of the trail radius
  - Can measure the echo's exponential decay time to estimate the ambipolar diffusion coefficient D_a
- The ambipolar diffusion coefficient relates to temperature and pressure: $D_a \propto \frac{T^2}{P}$
  - With care, we can estimate temperature from D_a



Verbeeck and Wislez [8]



Meteoroids and the meteor plasma itself

- Estimate mass flux from meteoroids (≈10 kg/s)
- Calculate meteoroid orbit and track populations
- Study meteor plasma dynamics and instabilities

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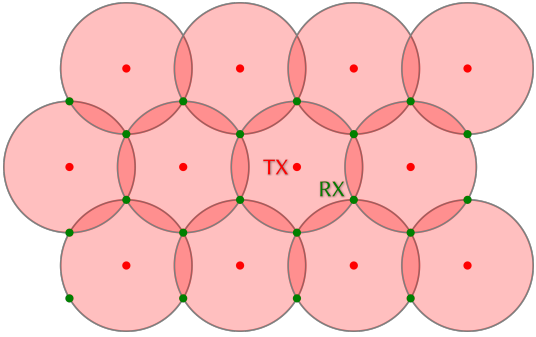
Multistatic meteor radar networks

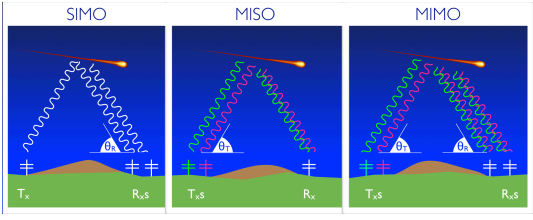
Coded CW technique

- Transmitters use continuous, independent pseudorandom codes
- Receivers decode signals from multiple transmitters
- Combinatorial number of links using one frequency

Advantages

- More detections (variety of specular geometries)
- No range aliasing
- Non-degenerate wind field sampling
- Flexibility in network configuration (SIMO vs MISO vs MIMO)



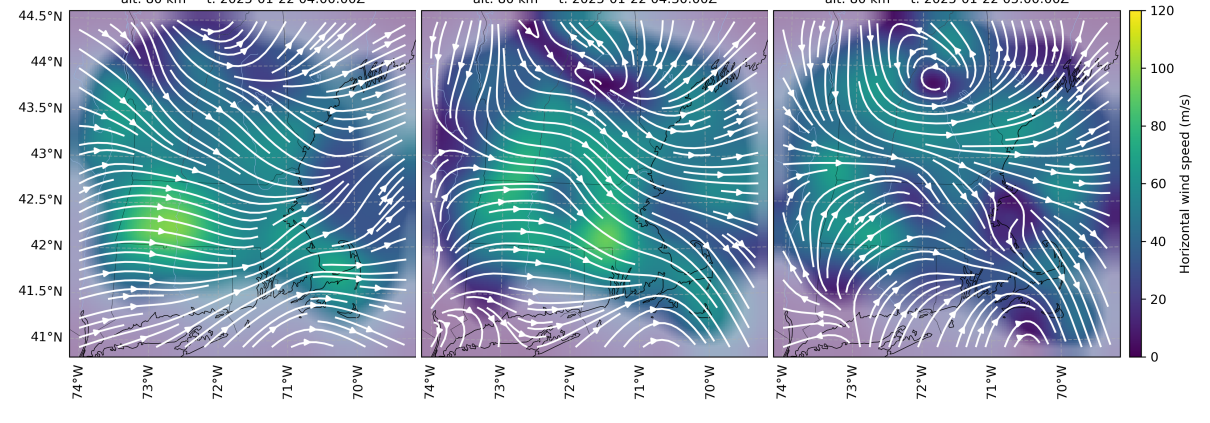


Chau et al. (2019). Novel specular meteor radar systems using coherent MIMO techniques to study the mesosphere and lower thermosphere. AMT, 12(4), 2113–2127. https://doi.org/10.5194/amt-12-2113-2019

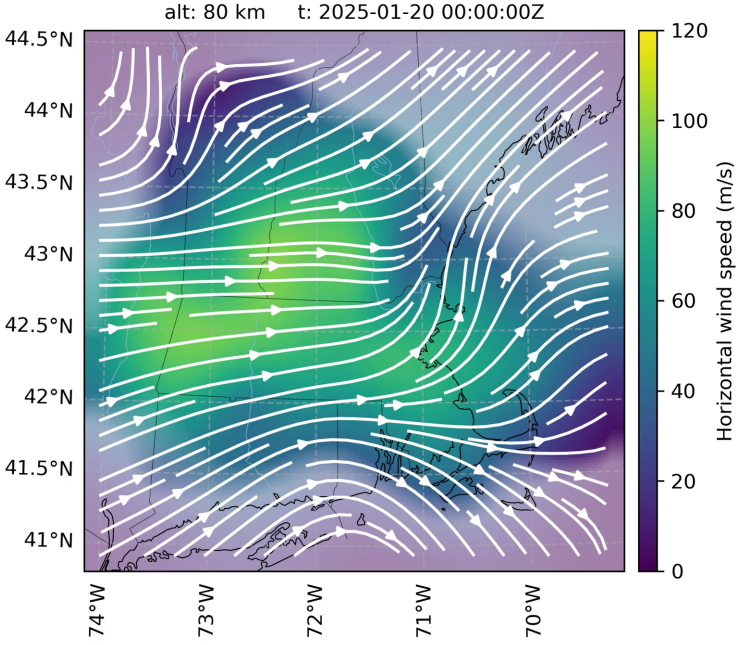
Wind fields

Gaussian process regression (GPR) wind field estimation

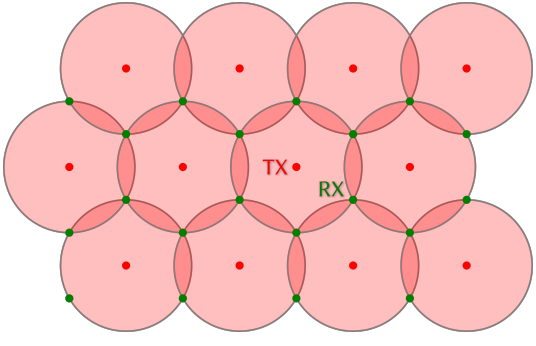
- Enough information to estimate 3-D wind field as a function of time
- Method assumes winds are locally correlated with $e^{-(d/l)^2}$ covariance kernel
  - Specified or fitted length scales for each spatial dimension and time



Example wind field movie - 80 km



Network scaling



- Each TX-RX link gives unique observations
- Key is to maximize the number of links with overlapping fields of view

Building out the network

- Backbone composed of TX array sites requiring more infrastructure
  - Where possible, co-locate with other instruments
  - Good for both science and access/maintenance
- Fill with single RX sites requiring less infrastructure
  - Engage with schools and amateurs for expansion

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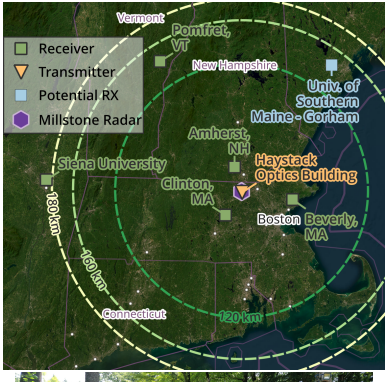
HamSCI and meteor radars

Avenues for engagement

- Bolster existing networks (SIMONe Haystack) with more receivers
  - GNSS-synchronized timing is critical!
  - What can be done with existing instrumentation (PSWS)?
- Open source meteor processing software
  - Extract science using uncontrolled transmitters?
  - Timing solution that doesn't rely on GNSS hardware?
- ... (your ideas)
- Anyone want to build a transmitter array? For fun?



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[1] M. D. Campbell-Brown and J. Jones. “Annual Variation of Sporadic Radar Meteor Rates”. In: Monthly Notices of the Royal Astronomical Society 367.2 (Apr. 2006), pp. 709–716. ISSN: 0035-8711. DOI: 10.1111/j.1365-2966.2005.09974.x.

[2] Zdeněk Ceplecha et al. “Meteor Phenomena and Bodies”. In: Space Science Reviews 84.3-4 (Sept. 1998), pp. 327–471. ISSN: 0038-6308, 1572-9672. DOI: 10.1023/A:1005069928850.

[3] Jorge L. Chau, Ronald F. Woodman, and Freddy Galindo. “Sporadic Meteor Sources as Observed by the Jicamarca High-Power Large-Aperture VHF Radar”. In: Icarus 188.1 (May 2007), pp. 162–174. ISSN: 0019-1035. DOI: 10.1016/j.icarus.2006.11.006.

[4] Lars Dyrud et al. “Plasma and Electromagnetic Simulations of Meteor Head Echo Radar Reflections”. In: Earth, Moon, and Planets 102.1-4 (June 2008), pp. 383–394. ISSN: 0167-9295, 1573-0794. DOI: 10.1007/s11038-007-9189-8.

[5] Gabrielle Guttormsen, Alex C. Fletcher, and Meers M. Oppenheim. “Atomic-Scale Simulations of Meteor Ablation”. In: Journal of Geophysical Research: Space Physics 125.9 (2020), e2020JA028229. ISSN: 2169-9402. DOI: 10.1029/2020JA028229.

[6] W. K. Hocking, B. Fuller, and B. Vandepeer. “Real-Time Determination of Meteor-Related Parameters Utilizing Modern Digital Technology”. In: Journal of Atmospheric and Solar-Terrestrial Physics. Radar Applications for Atmosphere and Ionosphere Research - PIERS 1999 63.2–3 (Jan. 2001), pp. 155–169. ISSN: 1364-6826. DOI: 10.1016/S1364-6826(00)00138-3.

[7] A. Pellinen-Wannberg. “The Radio Physics of Meteors: High Resolution Radar Methods Offering New Insights”. In: URSI Radio Science Bulletin 2011.339 (Dec. 2011), pp. 32–45. ISSN: 1024-4530. DOI: 10.23919/URSIRSB.2011.7909977.

[8] Cis Verbeeck and Jean-Marc Wislez, eds.
Proceedings of the Radio Meteor School, Oostmalle, Belgium, 10 - 14 September 2005. The International Meteor Organization, 2006. ISBN: 2-87355-017-1.

Appendix

Definitions: meteoroid, meteor, meteorite

Meteoroid

- Solid body in space
- Small (< 1 m in diameter)
- Fast (from 11 to 72 km/s)
- Exponentially more common as size decreases

Meteorite

- Body that survives passage through the atmosphere and lands on Earth

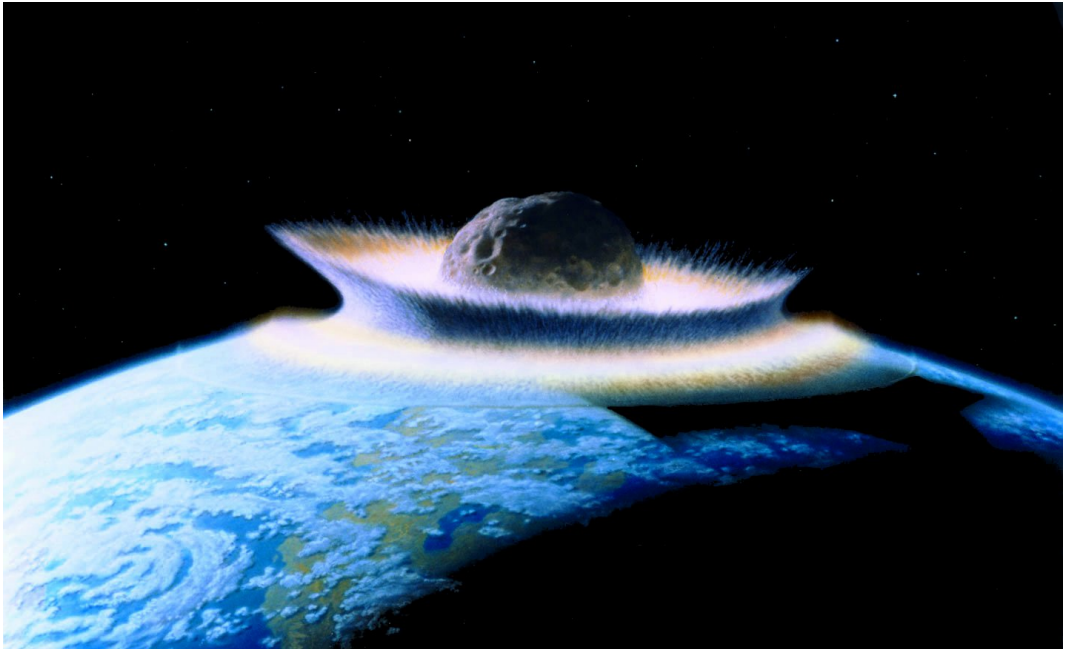
Micrometeorites
(landed on Earth)

Meteor

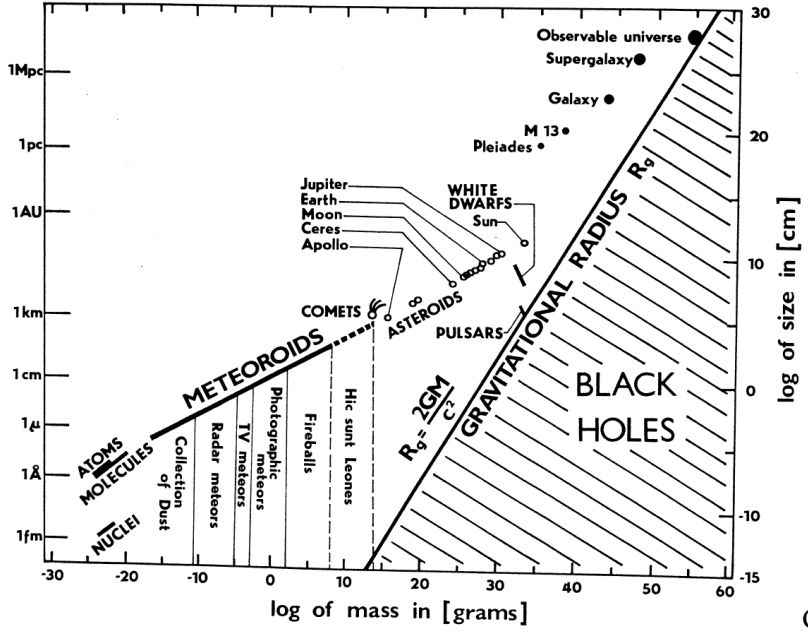
- Phenomena (light, plasma) caused by meteoroid entering the atmosphere
- Typically form at altitudes between 70 km and 120 km

NASA (2011)

Asteroid?



Size and mass of known objects



Ceplecha et al. [2]

Meteor showers

Showers

- Events with more meteors than normal
- Named for radiant (apparent source in the celestial frame)
- Associated with a specific parent body (occurrence corresponds to intersecting orbit)

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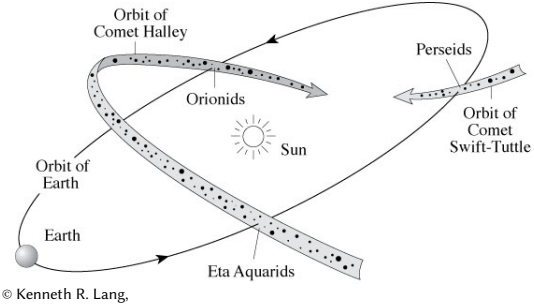


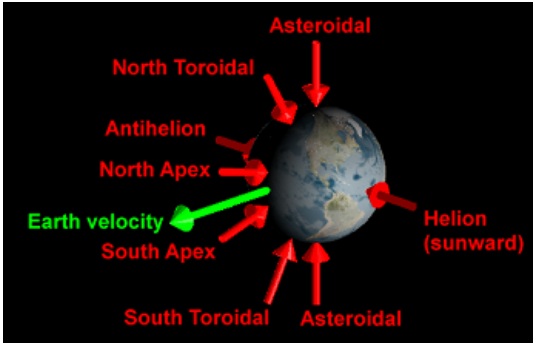
© Kenneth R. Lang,

https://ase.tufts.edu/cosmos/view_picture.asp?id=1305

Sporadic meteors

Sporadics

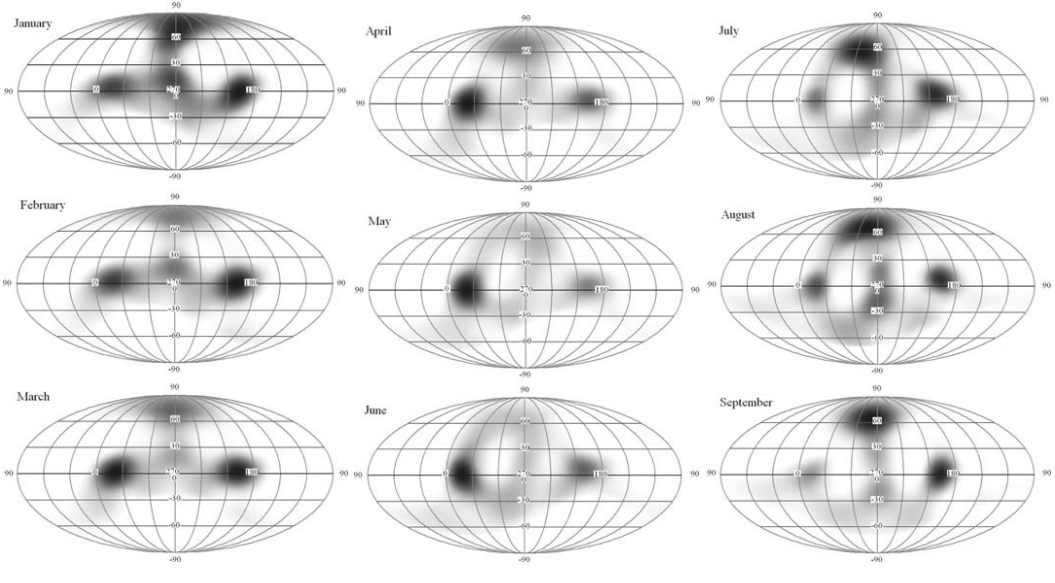
- Always occurring "background" events (not part of a shower)
- Grouped by radiant location with respect to Earth's orbit
  - Corresponding to different populations of Solar system objects
  - Radiant latitude is determined by the orbit inclination
  - Radiant longitude is determined by eccentricity, prograde/retrograde, inbound/outbound



Source	Direction	Eccentricity
North Apex	Retrograde	0.15–0.2
South Apex	Retrograde	0.15–0.2
Helion	Prograde	0.8–0.9
Anti-Helion	Prograde	0.8–0.9
North Toroidal	Prograde	0.15–0.2
South Toroidal	Prograde	0.15–0.2
Antapex	Prograde	0.2–0.3

Chau, Woodman, and Galindo [3]

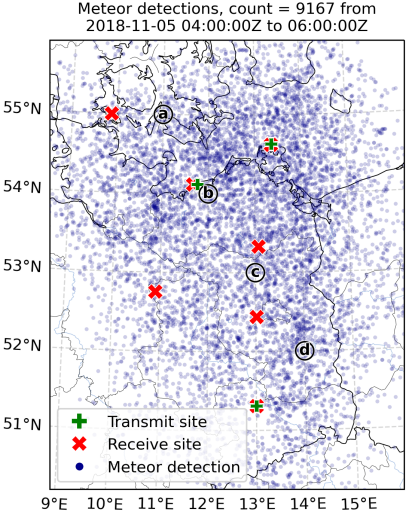
Sporadic meteor radiants



Campbell-Brown and Jones [1]

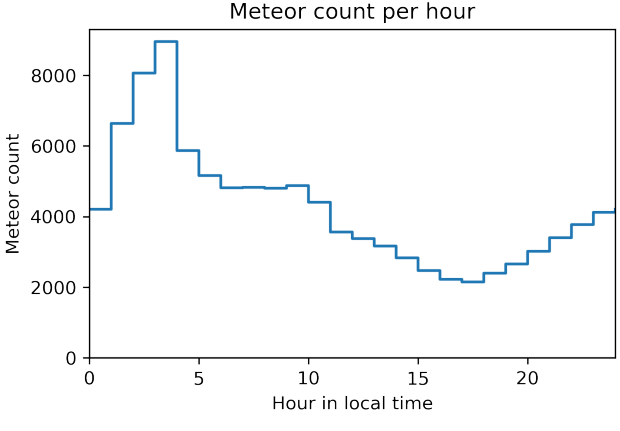
The random distribution of sporadic meteors

Spatial uniformity



Diurnal (and seasonal) patterns

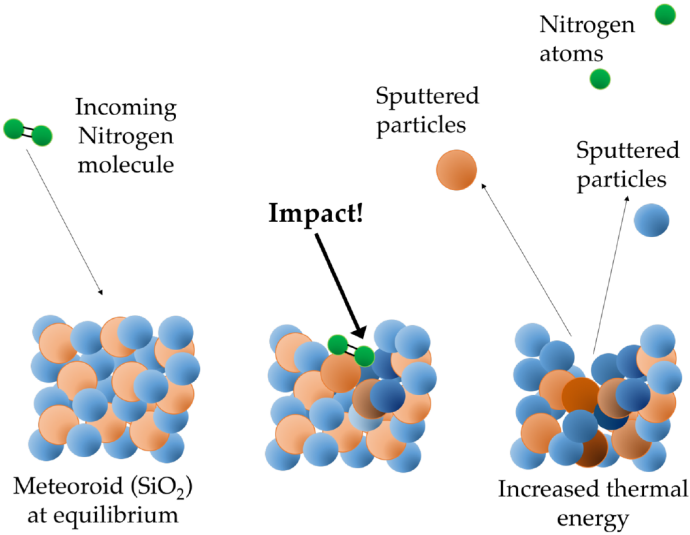
- Peak around sunrise



Meteoroid ablation

Ablation

- Meteoroid encounters increasing atmospheric density, heats up, and sheds particles

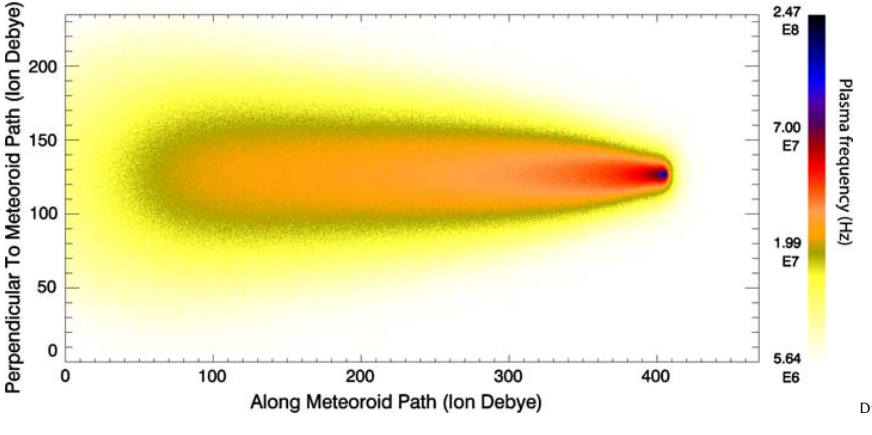


Guttormsen, Fletcher, and Oppenheim [5]

Meteor plasma

Plasma formation and evolution

1. Meteoroid particles and atmospheric gases collide and ionize, forming plasma
2. Dense plasma head surrounds meteoroid, diffusing plasma trail left in wake
3. Density eventually drops to background level (density $\leftrightarrow$ plasma frequency)



Dyrud et al. [4]